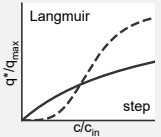
a

1 Ground truth

Verified non-isothermal column model (companion study)
Rate law: LDF (L1) or loading-dependent k(q) (L2)

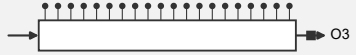


2 Design grid

Damköhler number Da: 0.1 to 1000, nine levels
Péclet number: about 470, 94, 19
Noise σ and isotherm error ε:
0, 0.5, 2 and 5 % each
60 replicates (15 select, 45 estimate)

3 Observation models

O1 probes of c, q, T
O2 probes of c, T (q rebuilt)
O3 outlet c(t), T(t) only



methods get a measured isotherm

4 Can the rate law be discovered?

Sparse regression (thresholding, ensemble, exact best subset), weak and strong forms; errors-in-variables (oracle); symbolic regression; neural network in the column model (demonstration); slow manifold; implicit regression without an isotherm (isothermal)
Success: exactly the true terms, signs, and q^* : $q = -1 \pm 10$ %

5 Is the rate constant identifiable?

The true law is given; only k is fitted, on the same observations with the same measured isotherm
Profile likelihood of k (O1 exact; O2 and O3 by full column solves)
Identifiable: 95 % interval inside [k/2, 2k]

6 Boundaries and the analyses that use them

H2b	discovery against identifiability boundary, 96 slices	confirmatory
H2a	does one error ratio place the boundary?	confirmatory
H2c	does errors-in-variables fitting move it?	confirmatory
	which methods discover the law, and where	descriptive
	choice among three candidate laws by likelihood	post hoc
	O2 and O3: probes without uptake, outlet only	descriptive
H2d	loading-dependent rate, step isotherm	exploratory

b

